B. TECH-ME-III-I Sem

23A03504e	NANO TECHNOLOGY (Professional Elective-I)	L	T	P	C
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COURSE OBJECTIVES

Course objectives: The objectives of the course are to	
1	**Understand the fundamentals of nano science and nanotechnology, including the history, classification and analyze the structural aspects of nanomaterials.**
2	Knowledge of the synthesis and fabrication techniques used in nano science, and methods for realizing semiconductor nanostructures.
3	Advanced characterization techniques used for analysing the structural, morphological, and electronic properties of nanomaterials.
4	Explore carbon nanomaterials properties and wide-ranging applications.
5	Familiarize with the diverse applications of nanotechnology, with emphasis on nanostructured thin films and quantum dots.

COURSE OUTCOMES On successful completion of this course the student will be able to		
CO1	Define and classify nanomaterials. Explain the historical development and scope of Nano science, and nanotechnology. Analyze the band structure and electronic behavior of nanomaterials.	L1, L2,L4
CO2	Explain the synthesis processes for bulk polycrystalline and single crystal materials. Differentiate between bottom-up and top-down fabrication approaches. Identify and select the requirements for semiconductor nanostructure fabrication and techniques	L2,L3,L5
CO3	Understand and explain the principles and applications of X-ray diffraction (XRD). Analyze optical properties of nanomaterials. Evaluate appropriate characterization technique.	L2,L3, L4,L5
CO4	Discuss and Characterize various carbon nanomaterials and its applications to analyze the types, synthesis methods, and Evaluate the impact of carbon nanomaterials in emerging technologies.	L2, L3, L4, L5
CO5	Identify and describe major applications, evaluate the impact in energy production, conversion, and environmental sustainability and analyze their applications and e**explore interdisciplinary applications of nano materials.**	L1, L2,L3, L4

UNIT-I

INTRODUCTION: History of nano science, definition of nano meter, nano materials, nano technology. Classification of nano materials. Crystal symmetries, crystal directions, crystal planes. Band structure.

PROPERTIES OF MATERIALS:

Mechanical properties, electrical properties, dielectric properties, thermal properties, magnetic properties, opto electronic properties. Effect of size reduction on properties, electronic structure of nano materials.

UNIT-II

SYNTHESIS AND FABRICATION: Synthesis of bulk polycrystalline samples, growth of single crystals. Synthesis techniques for preparation of nano particle – Bottom Up Approach – sol gel synthesis, hydro thermal growth, thin film growth, PVD and CVD; Top Down Approach – Ball milling, micro fabrication, lithography. Requirements for realizing semiconductor nano structures, growth techniques for nano structures.

UNIT-III

CHARECTERIZATION TECHNIQUES: X-Ray diffraction and Scherrer method, scanning electron microscopy, transmission electron microscopy, scanning probe microscopy, atomic force microscopy, piezoresponse microscopy, X-ray photoelectron spectroscopy, XANES and XAFS, angle resolved photoemission spectroscopy, diffuse reflectance spectra, photoluminescence spectra, Raman spectroscopy.

UNIT-IV

CARBON NANO TECHNOLOGY:

Characterization of carbon allotropes, synthesis of diamond – nucleation of diamond, growth and morphology. Applications of nano crystalling diamond films, grapheme, applications of carbon nano tubes.

UNIT-V

APPLICATIONS OF NANO TECHNOLOGY:

Applications in material science, biology and medicine, surface science, energy and environment. Applications of nano structured thin fins, applications of quantum dots.

TEXT BOOK:

Nano science and nano technology / M.S Ramachandra Rao, Shubra Singh/Wiley publishers.

Introduction to Nanotechnology by Risal Singh, Shipra Mital Gupta, Oxford Higher Education, First Publication 2016.

REFERENCE BOOKS:

Introduction to Nano Technology /Charles P. Poole, Jr., Frank J.Owens/Wiley publishers.

Nanotechnology /Jermy J Ramsden/Elsevier publishers (2015)

Nano Materials/A.K.Bandyopadhyay/ New Age

Nano The Essentials, T.Pradeep, McGrawHill, 2014

Nanotechnology the Science of Small / M.A Shah, K.A Shah/Wiley Publishers.

Online Learning Resources:

https://youtube.com/playlist?list=PLyqSpQzTE6M8682dGkNTN8936vSY4CbqZ&si=8S682KjXK7_xITpT

https://youtu.be/OLa8DQkKlyU?si=I6R1Of59MArQyPUb

https://youtu.be/u1ojNgPCHGs?si=mlIgQm4OdwZnHUo3